

AVATAR: Audio-Video Aligned Talking-head Animation with Recurrent Motion Prediction

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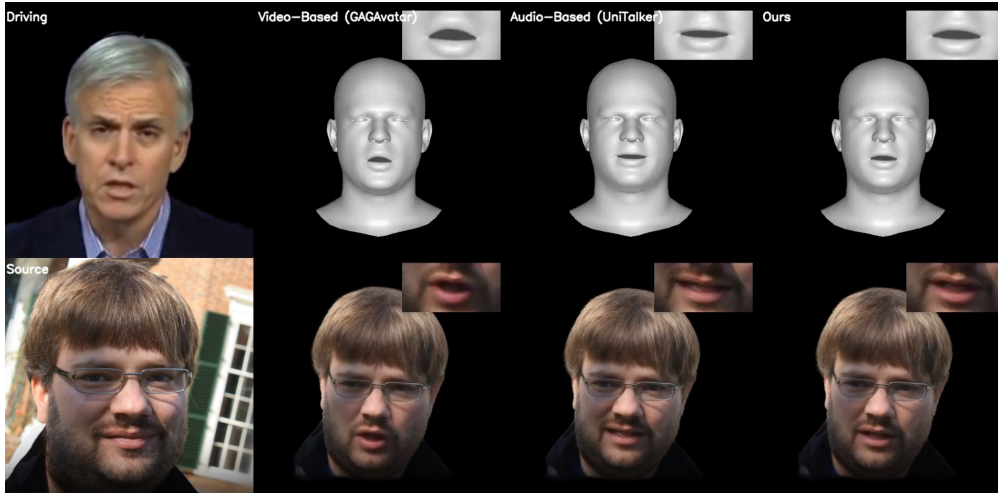


Figure 1. Our method generates more audio-synced talking heads while preserving the original video’s expressiveness.

Abstract

Existing methods for talking-head video generation often compromise between lip synchronization and natural head movements. Purely audio-driven methods ensure lip-sync accuracy but produce rigid, unnatural motions, while video-driven methods achieve expressive realism but struggle with accurate lip synchronization. To address this trade-off, we propose a multimodal framework that combines audio and video inputs. Our system uses pretrained UniTalker and GAGAvatar (3D face modeling modules) to initially predict facial motion from audio and extract ground-truth expressions and poses from driving videos. A Transformer-based refinement module, our sole trainable component, integrates these multimodal cues through cross-attention, effectively refining the predicted facial motion to ensure accurate lip synchronization and natural head movements. An Audio-Visual Speech Recognition (AVSR)-derived loss further enforces synchronization accuracy. Our approach helps enhance the realism and intelligibility of generated talking-head videos, bridging the

gap between audio-only and video-only methods. Demo website¹.

1. Introduction

Talking-head generation is the task of synthesizing a realistic video of a speaking face, given some driving input such as audio, video, or text. In essence, these methods animate a static face (e.g. a portrait image or 3D model) according to the content of a speech signal, a reference video’s motions, or a provided transcript. Approaches can be broadly categorized as video-driven, where facial movements are replicated from a driving video, and audio-driven, where lip movements and expressions are generated to match an audio track [14]. This technology has gained prominence for creating digital avatars and has broad applications ranging from virtual communication to entertainment and education. However, a core challenge in talking-head synthesis is achieving

¹https://morris8826.github.io/AVATAR_website/

both natural-looking head movements and perfectly synchronized lip motion with the speech [9].

Existing methods often face a trade-off between lip synchronization and motion realism. On one hand, video-driven techniques (often called face reenactment) capture the richness of real facial motion by transferring head pose and expressions from a reference video. These typically produce more life-like gestures and head movements; but if the driving video’s speech content differs from the target audio, the result is misaligned lip movements. In other words, the head motion may be natural, yet it is hard to manipulate the motion in existing videos to match the speech content [9], leading to poor lip-sync. On the other hand, audio-driven approaches use the input speech to directly generate or warp the mouth and face. This ensures that the mouth shapes correspond to the spoken phonemes, yielding accurate lip-sync. The downside is that without a visual driving signal, the generated talking head tends to have limited or repetitive motion – often a static head pose with minimal expression variation – resulting in rigid, unnatural animation. In short, audio-driven models excel at matching lips to speech, but can produce robotic or less expressive faces, whereas video-driven models inherit realistic dynamics at the cost of lip accuracy. Indeed, some recent one-shot talking head systems consciously trade a slight drop in lip-sync accuracy for much better head motion realism, a compromise that viewers prefer in user studies [25]. This dichotomy underlines the need for approaches that can simultaneously maintain accurate lip synchronization and natural movement.

To address these limitations, our project proposes a multimodal talking-head generation framework that combines both audio and video inputs to get the best of both worlds. By leveraging a driving video stream, the system can mimic genuine head poses and facial expressions from real footage, while the accompanying audio stream guides the fine lip movements to match the speech. In practice, the driving video (or an extracted motion prior) provides a baseline for global head motion and eye blinks, and the audio features ensure the mouth region moves in accordance with the spoken content. This dual input strategy allows the model to produce a talking face with synchronized lips and natural overall motion. Moreover, we introduce an auxiliary supervision signal derived from an Audio-Visual Speech Recognition (AVSR) model to explicitly enforce lip movement accuracy. Inspired by recent work that uses a pre-trained “lip-reading” expert to improve the visual intelligibility of generated speech [24], we incorporate a similar loss: the generated video’s audio-visual stream is fed into an AVSR network, and the model is penalized if the mouth movements do not correspond

to the correct spoken words. This auxiliary lip-reading loss provides direct feedback on articulation, ensuring that even subtle lip shapes align with the speech. We train a transformer-based model to fuse the audio and video streams and to generate refined facial motion parameters. The transformer’s sequence-to-sequence architecture encodes the temporal dependencies in speech and visual motion, effectively learning how to adjust the initial motions from the driving video to better match the audio. By using a transformer to model audio and video jointly (similar to approaches that encode audio synchronously with video, capturing long-range temporal context [24]), the system can produce smoother head movement and more precise mouth articulation. The result is an improvement in lip-sync fidelity without sacrificing the natural expressiveness provided by the video driver.

2. Related Work

2.1. Video-Driven Face Reenactment

Video-driven face reenactment aims to animate a target face image according to the expressions and head movements from a driving video. Early approaches often relied on 2D warping using facial landmarks or keypoints to transfer motion. For example, the First Order Motion Model (FOMM) of Siarohin et al. introduced a self-supervised framework that learns a set of keypoints and local affine transformations to represent motion, allowing a generator to warp the source image with the driving video’s motion patterns [21]. Such 2D warping methods (including extensions like MonkeyNet and thin-plate splines) preserve the source identity well for moderate movements. NVIDIA’s face-vid2vid extended this idea with a keypoint-based motion transfer and an identity-aware generator, achieving high-quality talking-head videos from a single source image and a driving video [26]. These approaches can produce convincing results, but they may struggle under large pose variations or extreme expressions, often requiring an inpainting network to handle occluded regions [21]. Purely generative (non-warping) methods can handle larger pose changes but sometimes sacrifice identity fidelity [28]. Recent works therefore explore hybrids that combine warping for coarse alignment with direct synthesis for fine details, aiming to get the best of both techniques [5].

An alternative line of research uses 3D parametric face models (such as 3D Morphable Models or FLAME) to drive reenactment. In these methods, one fits the driving video to a low-dimensional parameter space (encoding facial pose, expression, etc.), then applies those parameters to the target identity and renders the result. Classical graphics-based systems like Face2Face

[22] operate this way, fitting a 3D mesh to the source and target. More recent methods integrate 3D modeling with neural networks for greater photorealism. For instance, Thies et al. [23] propose Neural Voice Puppetry, which employs a latent 3D face model space to drive a deep generator; the underlying 3D representation provides temporal stability (preventing flicker) and structural consistency, while a neural rendering network produces realistic frames. Similarly, Kim et al.’s Deep Video Portraits first reconstruct 3D face parameters from video and then use a learning-based renderer to generate the output, transferring head pose, expression and even eye gaze in a coherent way [11]. By explicitly modeling 3D face geometry, these approaches handle large head rotations and expression changes more gracefully, although they require face model fitting and can be computationally heavier.

To achieve view consistency and high fidelity, researchers have also turned to neural rendering and Neural Radiance Fields (NeRF) for talking heads. NeRF-based reenactment methods represent the face as a continuous 3D field, enabling free viewpoint rendering and realistic lighting. Recent works have demonstrated dynamic radiance field models that learn a volumetric representation of a person’s head, producing highly photorealistic results and natural pose control [20]. For example, models like DFRF and NerFACE learn dynamic neural radiance fields that can animate a portrait with 3D consistency [8]. These yield impressive realism (even allowing one-shot or few-shot reenactment with correct 3D perspective), but often require abundant training data of the person or sophisticated optimization, making them less flexible for arbitrary subjects.

While these approaches can produce photorealistic frames, ensuring perfect lip synchronization in speech scenarios remained a challenge without using the audio information. To address the lip-sync limitation, some video-driven methods began to incorporate audio cues alongside the driving video. The recently proposed Audio-Visual Face Reenactment GAN (AVFR-GAN) is one such example: it animates a source image using a driving video for head motion (via a dense motion field from learned keypoints) and uses the accompanying audio to guide the mouth movements [1]. By providing the speech content as input, the network can focus on generating accurate lip shapes, resulting in much improved lip synchronization. The authors report that adding audio helps the model attend to the mouth region and yields clearly better lip-sync without sacrificing the head pose and expression realism from the driving video. Earlier, Zhou et al. [31] took a step toward multimodality with their adversarially disentangled audio-visual representation learning (DAVS). They disentangle identity and speech content, allowing a talking face to be driven by

either a sample video or an audio clip in a unified framework. This two-stream design demonstrated that combining audio and video information can produce more accurate and expressive reenactments than using video alone.

2.2. Audio-Driven Face Reenactment

Audio-driven talking head generation focuses on producing a realistic talking face video given only a speech audio track (and usually a static image of the target person). Early works in this area often used a two-step approach: first mapping audio to intermediate representations like phoneme sequences or lip motion parameters, then generating or retrieving video frames to match those parameters. For instance, classic methods like Video Rewrite [2] concatenated mouth images based on phoneme recognition. With deep learning, researchers moved toward direct audio-to-video mappings. Chung et al. and others trained convolutional networks to predict a sequence of face images from audio, initially limited to a specific speaker or with simple mouth animations. As the field progressed, the aim became to generalize to arbitrary subjects. X2Face [27] could drive a new face from audio by embedding the audio into the same latent space used for image-driven motion, demonstrating early on that a single model can handle both audio and video driving signals. Subsequent models like ATVGnet [3] improved the quality and generalization, using encoder-decoder architectures to generate the lower half of the face that matches the speech. These models produced recognizable lip movements, but often the head pose was fixed and the expressions were minimal, since the audio alone does not provide those cues.

In recent years, GAN-based and network-based approaches have significantly advanced audio-driven reenactment. LipGAN [12] was one of the first GAN models for general-purpose lip syncing, able to paste synchronized lip movements onto a target face. It showed that a conditional GAN could generate mouth movements that align with input speech for many identities. Building on this, Wav2Lip [17] introduced a powerful lip-sync discriminator (a pre-trained expert model) to enforce near-perfect synchronization during training. As a result, Wav2Lip achieved lip-sync accuracy on par with real videos – it is speaker-independent and language-agnostic, able to make any talking face video match a new audio track with only a single frame of that person as reference. This represented a leap in lip-sync quality: human evaluators largely preferred Wav2Lip outputs in terms of sync fidelity. However, these audio-driven methods typically focus on the mouth region. They assume a mostly static head pose and neutral expression, which can lead to somewhat stiff or less expressive results. For example, an audio-driven model

might generate correct lip movements but the head and eyes of the character remain unnaturally frozen. Indeed, Chen et al. [29] observe that earlier audio-driven systems like ATVGnet, DAVS and others produced talking faces with no head motion or eye blinking, resulting in less natural videos. This lack of natural motion and expressiveness is a known limitation when using only audio input.

To make audio-driven talking heads more life-like, researchers have explored adding temporal modeling and expressive variations. Some works predict not just each frame’s mouth shape but also the trajectory of head pose or facial expressions over time. For instance, MakeItTalk [33] introduced a model that learns a speaker-specific style code and uses an LSTM to generate rhythmic head movements and eyebrow motions in sync with speech, thereby avoiding a static, “puppet-like” effect. Another strategy is predicting intermediate motion fields from audio: e.g., using a neural network to infer dense optical flow or keypoint displacements that the image should undergo, and then warping a source image accordingly. This can create smoother, more continuous motion than predicting frames outright. Recent transformer-based models have also shown promise. FaceFormer [7] is a speech-driven 3D facial animation model that uses a Transformer to capture long-term audio context, rather than just short phonetic segments. By encoding longer audio sequences, it avoids some of the occasional mismatches in lip shape that occur with short context windows. The transformer learns the alignment between audio and facial motion over an extended period, producing more accurate and contextually appropriate mouth movements (including coarticulation effects) and more stable predictions for jaw and cheeks. Such temporal and contextual modeling helps improve the naturalness and intelligibility of the generated speech movements. Likewise, some approaches integrate pre-trained speech representations (e.g. HuBERT or wav2vec features) to better drive the visuals, or diffusion models to enhance video quality. Despite these improvements, a common issue remains: limited expressiveness. Most audio-only models default to a neutral expression (apart from lip and maybe slight jaw movement) and do not capture the full range of emotional or idiosyncratic facial gestures a real person would display when speaking.

To overcome the above limitations, combining audio with additional visual inputs or controls has proven effective. One notable work is PC-AVS (Pose-Controllable Audio-Visual System) [32] explicitly addresses the problem of frozen head pose in audio-driven generation. PC-AVS introduces a learned pose code alongside the audio features, effectively modularizing the audio-visual representation. In practice, this means

the system can take a separate driving signal for head motion (for example, a reference video or a parametric code) in addition to audio, and generate a talking head where the lips follow the audio and the head moves following the pose input. By disentangling “what is being said” from “how the head is moving,” they achieve rhythmic head movements and even allow head pose control without needing explicit 3D landmarks. This results in more natural and personalized animations — the character can nod, turn, or tilt their head while speaking, which audio alone cannot induce. Similarly, some audio-driven systems incorporate random subtle head motion or blinks to avoid dead-eye stares. The trend in recent work, such as AVFR-GAN and others, is toward audio-visual fusion: leveraging the strength of audio (accurate lip sync) together with a secondary visual modality (another video, a pose prior, or a driving frame) to enrich the motion. These multi-modal systems outperform single-modality models in both lip synchronization and overall realism [1, 31], as the audio ensures the speech is matched while the visual input provides cues for pose, timing, and expression dynamics that audio alone might miss.

Our approach. Given the strengths and weaknesses of purely video-driven and purely audio-driven reenactment, our work falls into this emerging multimodal category. We propose a transformer-based model that leverages both audio and video inputs simultaneously to generate talking-head videos. By using the driving video, our model can mimic the head pose, gaze, and general facial expressions of the target performance, maintaining realism and expressiveness. At the same time, the audio input guides the fine-grained mouth shapes, ensuring precise and robust lip synchronization. To further improve the articulation of speech, we introduce an auxiliary Audio-Visual Speech Recognition (AVSR) loss during training. In essence, we have a pre-trained AVSR model “watch and listen” to the generated video and encourage the network to produce outputs that the AVSR can transcribe correctly. We describe our method in details in the next section.

3. Methodology

Our talking-head generation system is a multimodal pipeline that takes (1) a driving speech audio (and its corresponding real video for training), and (2) a static source image of a target person, and produces a synthetic video of the target face speaking the given audio. The overall architecture, as illustrated in Figure 2, integrates several pre-trained components with a small trainable module. We adopt the FLAME 3D face model parameters to represent facial motion, where each video frame’s facial state is described by: a 300-dimensional

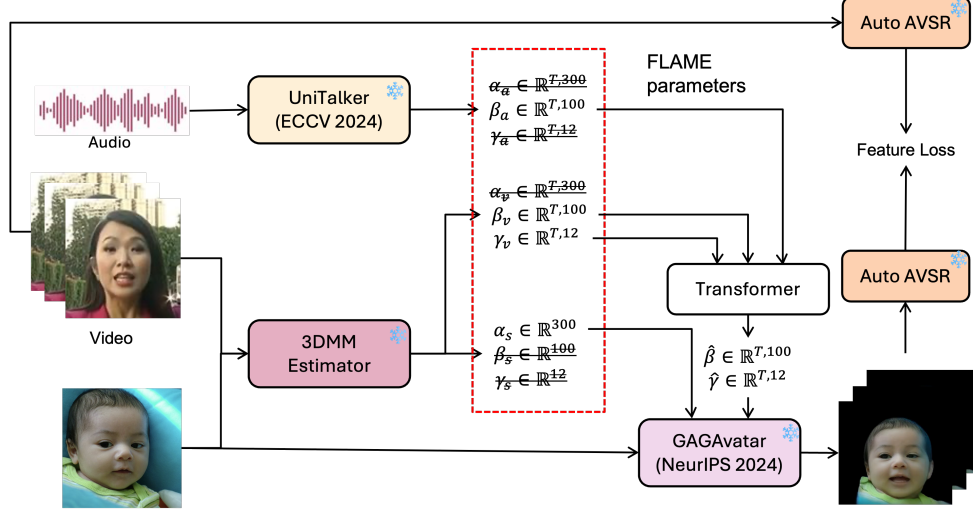


Figure 2. Overview of our lip-synchronized talking-head generation system. Given a target speaker image and a driving video with accompanying audio, our system synthesizes a photorealistic talking-head video that closely matches the speech content and visual articulation. The driving audio is first processed by UniTalker (pretrained) to produce audio-driven expression coefficients $\beta_a^{(t)}$. Simultaneously, a 3DMM Estimator extracts the ground-truth expression $\beta_v^{(t)}$ and pose $\gamma_v^{(t)}$ parameters from the driving video. These two modalities are fused by a Transformer-based refinement module (the only trainable component), which uses cross-attention between audio- and video-derived parameters to predict refined coefficients ($\hat{\beta}^{(t)}, \hat{\gamma}^{(t)}$). These refined coefficients, along with identity parameters from the target image, are fed into GAGAvatar to render the output video frames. To ensure high-quality lip synchronization, we supervise training using a pretrained AVSR model that provides an embedding-space loss by comparing representations of the predicted video and the ground-truth driving video.

identity shape vector α , a 100-dimensional expression coefficient vector β , and a 12-dimensional pose parameter vector γ (for head rotation, jaw, etc.). For clarity, we use subscripts to denote the source of these parameters: for example, β_a denotes expression from audio, β_v from the driving video, α_s from the source image, and $\hat{\beta}$ for a refined/predicted parameter. The key stages of the pipeline are as follows:

Audio-driven Coefficient Prediction. The input audio is processed by a pre-trained UniTalker model [6], which generates an initial sequence of facial motion coefficients ($\beta_a^{(t)}, \gamma_a^{(t)}$) for each time frame t aligned with the audio. These represent the facial expressions (notably lip and jaw movements) and coarse head pose that UniTalker predicts purely from the speech signal. UniTalker was trained on large audio–video datasets to map speech to realistic facial motion, so it provides a strong audio-driven prior for lip-sync and some head motion.

Visual Coefficient Extraction. In parallel, the driving video frames are processed by a 3DMM estimator (e.g. a FLAME fitting network) to extract the ground-truth 3D face coefficients ($\alpha_v^{(t)}, \beta_v^{(t)}, \gamma_v^{(t)}$) for each frame t . This provides the true facial expression and head pose trajectories from the real video performance. We also extract the static source image’s FLAME parameters ($\alpha_s, \beta_s, \gamma_s$) using the same esti-

imator. Here $\alpha_s \in \mathbb{R}^{300}$ encodes the target person’s identity-specific shape. In our reenactment setting, we retain α_s as the identity of the avatar, while the expression β_s and pose γ_s from the single image may be used to initialize the avatar (typically the source is a neutral, frontal face). All these parameters are represented in the common FLAME space, allowing us to transfer expressions and poses from one identity to another.

Motion Refinement Transformer. The Transformer-based refinement module is the only trainable component in our system. It is designed to refine the expression and pose parameters by combining audio-derived and video-derived motion cues. Specifically, at each time step t , the model takes as input the expression coefficients from the audio $\beta_a^{(t)}$, and both the expression $\beta_v^{(t)}$ and pose $\gamma_v^{(t)}$ coefficients from the driving video. The goal is to predict a refined expression and pose output ($\hat{\beta}^{(t)}, \hat{\gamma}^{(t)}$) that improves upon the audio-only prediction, ensuring accurate lip synchronization and natural head motion. We construct two sequences: the audio sequence $\beta_a^{(t)}$ serves as the query, and the video sequence $[\beta_v^{(t)} | \gamma_v^{(t)}]$ serves as the key and value for cross-attention. At each time step, the Transformer computes attention between the audio expression and the corresponding video-derived motion to selectively fuse relevant cues. The attended outputs



Figure 3. Qualitative comparison of mouth synchronization with the audio and expressiveness.

are then passed through a lightweight Transformer encoder (4 layers, 4 head, 256-dimensional hidden states) with temporal self-attention, which captures dependencies across time and produces the refined outputs $(\hat{\beta}^{(t)}, \hat{\gamma}^{(t)})$. This architecture allows the model to preserve the speech-driven dynamics from audio while grounding its predictions in the realistic temporal patterns of head and facial motion from the video. The refinement process enables the system to correct potential mismatches in lip shape or missing head gestures, enhancing the expressiveness and intelligibility of the generated talking head. This approach extends prior work such as SadTalker [30], with the novel integration of an AVSR-guided supervision that further enforces phonetic accuracy.

Avatar Rendering with GAGAvatar. The pre-trained GAGAvatar engine (Generalizable and Animatable Gaussian Avatar) is used to render the actual video frames of the talking head [4]. GAGAvatar is a one-shot neural head avatar model that can generate a 3D representation of a person’s head from a single image and render it under novel expressions and poses in real time. We first construct the 3D Gaussian head avatar for the source identity using the source image (this produces a neural volumetric representation that encodes α_s , the person’s identity and appearance). Then, for each frame t , we drive this avatar using the refined coefficients $(\hat{\beta}^{(t)}, \hat{\gamma}^{(t)})$ to synthesize a photorealistic output frame. In essence, GAGAvatar decodes the target’s identity shape α_s together with the time-varying expression and pose into an image of the talking face. Because GAGAvatar is generalizable to unseen identities and operates efficiently via a 3D Gaussian splatting technique, it can animate the new face with high fidelity and without per-subject train-

ing. This yields the final output video of the target face speaking.

AVSR Loss. To ensure the generated video’s lip movements convey the correct speech content, we employ a pre-trained Audio-Visual Speech Recognition (AVSR) model in our pipeline. We feed both the real driving video and the generated video into a pre-trained AVSR network to generate corresponding encoder features and then apply a mean-squared error loss between the two representations. The AVSR model is a state-of-the-art conformer-based speech recognizer that leverages and lip movements; notably, Auto-AVSR achieves near-human-level word recognition error rates ($\approx 0.9\%$ WER on the LRS3 dataset) by training on large-scale AV speech data. During training, we compare the AVSR feature sequences for the real vs. generated videos and penalize any differences. This AVSR-guided loss encourages the generated video to match the true lip movements, effectively ensuring that an observer (the AVSR) “lip-reads” the same phrase from the synthetic video as from the real video. By using the AVSR’s intermediate feature representations, we obtain a differentiable supervision signal that enforces correct mouth movements without needing ground-truth target video of the new identity. This idea is similar to the lip-reading loss used in prior talking head works, but here we leverage a full audio-visual speech recognizer for robust supervision.

4. Experiments

4.1. Experiment Setting

Datasets We use videos from FaceForensics++ [19] as driving inputs and sample source face images from FFHQ [10]. Since the Auto AVSR model [15] we adopt

only supports English, we apply a pre-trained language classifier [18] to retain English-speaking videos. Valid source-driving pairs are then randomly assigned. All videos are re-sampled to a consistent frame rate of 30 fps for synchronization.

As the length of FLAME parameters generated by the audio-based method [6] may not match the number of video frames, we align them to the video’s frame count. Let T_a and T_v denote the number of FLAME parameter frames from audio and video, respectively. If $T_a > T_v$, we truncate the audio-based parameters to the first T_v frames; otherwise, we pad them by repeating the last available frame until reaching T_v .

Implementation details Our framework is implemented in PyTorch and utilizes FLAME [13, 16] as the underlying 3DMM model. We extract FLAME parameters from the driving video following the approach of [4], while the audio-driven FLAME parameters are obtained using UniTalker [6]. To train the model, we adopt a self-reenactment strategy, where the first frame of the driving video serves as the source image. We randomly sample 1.2 seconds from the driving video and extract embeddings using the Auto AVSR encoder [15]. Training is performed for 1,000 iterations with a total batch size of 8 on two NVIDIA RTX 4090 GPUs. All model components are frozen except for the transformer module, which is trained to refine the expression and pose parameters.

4.2. Results

Qualitative results The results in Figure 3 demonstrate the effectiveness of our approach in synthesizing realistic and expressive talking-head videos. Each row illustrates the generated frames over time for different target identities driven by the same source video and audio input. Our model produces coherent and visually natural facial movements, accurately capturing lip synchronization and subtle expressions. Notably, the close-up lip insets highlight our method’s precise alignment with the speech content, ensuring phonetic consistency. Overall, these results show that our Transformer-based refinement strategy, combined with multimodal input and AVSR supervision, improves lip movements while retaining high-quality talking-head outputs with naturalistic expressions (see our demo page for better visualization).

5. Limitations & Future works

Our proposed method significantly enhances lip synchronization in talking-head videos by leveraging AVSR-guided refinement. However, several shortcomings remain. Notably, we observed inconsistencies be-

tween refined FLAME parameters and the final rendered images: even though the underlying mesh correctly indicates open mouth shapes, the rendering process frequently results in lips appearing separated without displaying teeth, leading to unnatural visual artifacts. See Figure 4. Additionally, the overall quality heavily depends on accurate initial predictions from pretrained models; errors from UniTalker or the 3DMM estimator propagate downstream, limiting corrective capabilities.



Figure 4. The rendered image still fills in the teeth region completely, even though the underlying mesh indicates an open mouth with visible gaps. This inconsistency reduces the realism of the synthesized output.

Although we see improvement in performance qualitatively, we need to perform more thorough quantitative experiments like word error rate calculation to establish the improved performance. Future work should explore joint end-to-end training of the entire pipeline, allowing the transformer refinement module and the rendering model to directly inform each other, thus reducing dependency on pretrained modules and potentially correcting errors more effectively. Additionally, incorporating explicit constraints or perceptual losses (such as GAN-based adversarial loss or facial landmark consistency) could help resolve rendering inconsistencies, especially those affecting mouth and teeth visibility, enhancing realism. Finally, adopting autoregressive generation strategies would address scenarios with mismatched audio-video durations, enabling the system to reliably synthesize longer, diverse, and more robust talking-head sequences suitable for various real-world applications.

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